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Tutorial - 4

(Logistic Regression)

An email service provider wants to classify emails as spam or not spam using logistic regression. Data from 10 emails has been collected.

Suspicious words (x_1)	External links (x_2)	Attachment (x_3)	Spam
5	2	1	1
1	0	0	0
3	1	0	1
0	0	0	0

w_1 w_2 w_3

w_0 - Bias = 0.

Let $w_1 = 1$

$w_2 = 1$

$w_3 = 1$

$\hat{y}$ Predicted	z
0.99	8
0.78	1
0.98	4
0.5	0

$z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$

$z_1 = 0 + 1(5) + 1(2) + 1(1)$

$z = 8$

$z_2 = 0 + 1(1) + 1(0) + 1(0)$

$z_2 = 1$

$z_3 = 0 + 1(3) + 1(1) + 1(0)$

$z_3 = 4$

$z_4 = 0 + 1(0) + 1(0) + 1(0)$

$z_4 = 0$

C.F (J)
$y - \hat{y}_{pre}$
0.01
-0.78
0.02
-0.5

$$\hat{y}_{pre} = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$\text{Cost Function} = J = \frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

$$\frac{\partial J}{\partial w} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)\,x_i$$

$$w_0^{new} = w_0^{old} + \eta\left(\frac{\partial J}{\partial w}\right)$$

$$= w_0^{old} + \eta\left(\frac{1}{N}\sum_i (y_i - \hat{y}_i)\right)$$

$\boxed{N = 4}$

$$w_0^{new} = 0 +$$

$$\begin{array}{r} 1.28 \\ 0.03 \\ \hline 1.25 \end{array}$$

$$\frac{\partial J}{\partial w} = \frac{1}{N}\sum_{i=1}^{N} (y_i - \hat{y}_i)$$

$$= \frac{1}{4}\left((0.01) + (-0.78) + (0.02) + (-0.5)\right)$$

$$= \frac{1}{4}\left[0.01 - 0.78 + 0.02 - 0.5\right]$$

$$\begin{array}{r} 0.01 \\ 0.02 \\ \hline 0.03 \end{array}$$

$$= \frac{1}{4}\left[0.03 - 1.28\right]$$

$$\begin{array}{r} -0.78 \\ -0.5 \\ \hline 1.28 \end{array}$$

$$= -\frac{1.25}{4}$$

$$= -0.3125$$

<u>weights</u>

$$w_j^{new} = w_j^{old} + \eta\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)\cdot x_j$$

$$= w_j^{old} + \eta\left(\frac{\partial J}{\partial w}\right)\cdot x_j$$

$$w_1^{new} = 1 + \frac{1}{4}\left[((0.01)\times 5) + (-0.78\times 1) + (0.02\times 3) + (-0.5\times 0)\right]$$

$$= 1 + \frac{1}{4}\left[0.05 - 0.78 + 0.06 - 0\right]$$

$$= 1 + \frac{1}{4}\left[-0.67\right]$$

$$= 1 - \frac{0.67}{4}$$

$$= 1 - 0.1675$$

$$= 0.8325$$

$$w_2^{new} = 1 + \frac{1}{4}\left[(0.01\times 2) + (-0.78\times 0) + (0.02\times 1) + (-0.5\times 0)\right]$$

$$= 1 + \frac{1}{4}\left[0.02 - 0 + 0.02 - 0\right]$$

$$= 1 + \frac{1}{4}\left[0.04\right] = 1 + \frac{0.04}{4} = 1 + 0.01 = 1.01$$

$w_3^{new} = w_3^{old}$ $1 + \frac{1}{4}\left[(0.01 \times 1) + (-0.78 \times 0) + (0.02 \times 0) + (-0.5 \times 0)\right]$

$= 1 + \frac{1}{4}\left[0.01 - 0 + 0 + 0\right]$

$= 1 + \frac{0.01}{4}$

$= 1 + 0.0025$

$= 1.0025$

$Z_{New} = w_0^{new} + w_1^{new} x_1 + w_2^{new} x_2 + w_3^{new} x_3$

$Z_1^{new} = w_0^{new} + w_1^{new} x_1 + w_2^{new} x_2 + w_3^{new} x_3$

$Z_{1,New} = -0.3125 + 0.8325(5) + 1.01(2) + (1.0025)(1)$

$Z_1^{new} = 6.8725$

$Z_{2New} = -0.3125 + 0.8325(1) + 1.01(0) + (1.0025)(0)$

$Z_2 = 0.52$

$Z_{3\,new} = -0.3125 + 0.8325(3) + 1.01(1) + (1.0025)(0)$

$= 3.195$

$Z_{4New} = -0.3125 + 0.8325(0) + 1.01(0) + (1.0025)(0)$

$= -0.3125$

$y^n = \frac{1}{1+e^{-z}}$

$y_1^n = 0.999$ $y_1^n = 1 \rightarrow$ correct prediction

$y_2^n = 0.6272$ $y_2^n = 1 \rightarrow$ not correct prediction

$y_3^n = 0.961$ $y_3^n = 1 \rightarrow$ corret prediction

$y_4^n = 0.4226$ $y_4^n = 0 \rightarrow$ correct prediction.